Fiscal Year:	FY 2022	**Task Last Updated:**	FY 12/23/2022
PI Name:	Bershad, Eric M. M.D.		
Project Title:	Enhancing eXploration Platforms and ANalog Definition Intracranial Pressure (EXPAND-ICP)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77030-3411	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	TRISH--EXPAND
Start Date:	05/01/2022	**End Date:**	04/30/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	SpaceX		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hirzallah, Mohammed M.D. (Baylor College of Medicine) Williams, Michael MD (University of Washington) Malm, Jan Ph.D. (University of Umea, Sweden) Eklund, Anders Ph.D. (University of Umea, Sweden)		
Grant/Contract No.:	NNX16AO69A-EXP0013		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This is an EXPAND Project. In this proposal, we aim to measure the intracranial pressure (ICP) before and after spaceflight, using the well-established lumbar puncture method in commercial spaceflight participants on an upcoming SPACE-X mission. We will use methods as similar as possible to those used in the Direct ICP protocol so that ICP measurements can be compared between the studies. The advantage of this proposal for obtaining an invasive ICP is that the criteria for being selected to a commercial spaceflight mission are much less rigorous, compared to the NASA standards; and therefore, it is highly unlikely that an incidentally elevated pre-flight ICP would disqualify the participant from the mission. Secondly, most NASA astronauts have engaged in a variety of extreme physical feats, including centrifugation, high altitude ascent (i.e., mountain climbing), sky diving, parabolic flight, and high levels of acceleration/high G aircraft maneuvers, which may have an effect on baseline ICP. Therefore, an understanding of the effect of the microgravity environment on ICP in individuals who more closely represent that of the general population is valuable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2022.
Bibliography Type:	Description: (Last Updated: 06/17/2026)